ESPaperPlay / DEVELOPMENT BOARD r0

Two editable power sheets. ESP32-S3-WROOM-1-N16R8 / GDEY075T7-T01.

DESIGN REVIEW ONLY: not a full board, not a manufacturing release.

POWER_INPUT_CHARGER



文件: POWER_INPUT_CHARGER.kicad_sch

POWER_3V3_CONTROL



文件: POWER_3V3_CONTROL.kicad_sch

DEFINED IN THIS DRAFT

BQ24074 power path; protected battery / NTC interface; default charge inhibit.
TPS63802 3.3 V regulator; measurement isolation; TCA9539 reset and control.
MAX17048 battery gauge; AON-safe open-drain VBUS detector.
Cross-sheet global nets connect the two sheets; no hidden MCU connections.

ASSEMBLY AND INTEGRATION GATES

R108 CHG_ARM: DNP until the actual cell and NTC temperature window are approved.
J105 RECOVERY_CE: normally open. J202 MAIN_RUN: one shunt, 1-2 RUN / 2-3 OFF.
USB-C protection, MCU, domain load switches, touch isolation and audio are not drawn yet.
Battery connector pinout, converter DLA footprint, power-link current ratings remain TBD.
ERC checks schematic rules; it does not validate charging, thermal behavior or removable links.

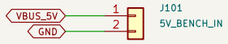
Design contract: docs/v2/hardware/power-design-spec.md
Review instructions and exact release limits: hardware/devboard-r0/README.md

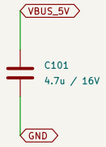
DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa

Sheet: /		
File: ESPaperPlay_Devboard.kicad_sch		
Title: ESPaperPlay DEVELOPMENT BOARD - POWER DRAFT		
Size: A3	Date: 2026-11-01	**Rev: r0 DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/3

1 POWER PATH AND PROTECTED BATTERY INTERFACE

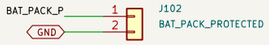
5 V only. USB-C / ESD / input protection belong to the future MCU_USB sheet.

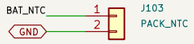




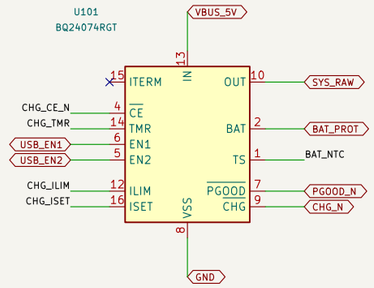
Known, current-limited source only.
Connector / protection BOM not released.

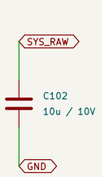


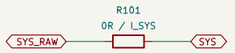


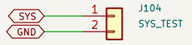


BAT + NTC are functional bench terminals.
Pack connector / polarity / R-T curve TBD.
No fixed resistor in place of pack NTC.





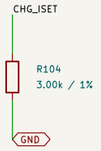




R101: measure total SYS load.
CE pull-up stays on SYS_RAW side.









ISET / ILIM / TMR = calculation starting points, not approved charging settings.
ITERM pin 15 NC: internal termination threshold (mode dependent).

3 AON-SAFE VBUS SENSE



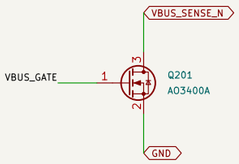


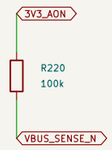




PWR_FLAG: normal RUN with R101/R203 populated.
ERC does not model removable links or input protection.



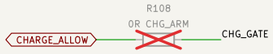


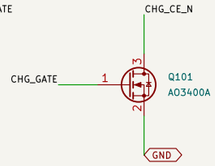




LOW = VBUS present. AON-off isolation.
PGOOD still required for valid input.

2 HARDWARE CHARGE INHIBIT / CURRENT MODE / STATUS







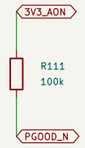








EN2 EN1: 00=100mA, 01=500mA,
10=RILIM, 11=USB suspend.
Do not infer source capacity from VBUS.





J105 normally OPEN. Recovery only: MAIN_RUN OFF, AMP disabled,
EN1/EN2=00, qualified pack + NTC, known bench source.
R108 is DNP: firmware cannot enable charging in initial assembly.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

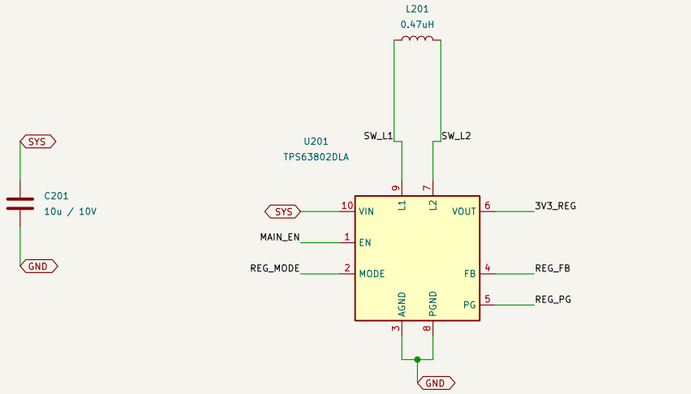
ESPaperPlay / xiaobaoaaaaaa

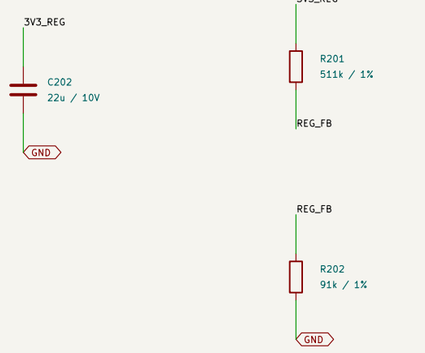
Sheet: /POWER_INPUT_CHARGER/
File: POWER_INPUT_CHARGER.kicad_sch

Title: POWER INPUT / BQ24074 CHARGER

Size: A3	Date: 2026-11-01	**Rev: r0 DRAFT**
KiCad E.D.A. 10.0.7		Id: 2/3

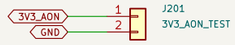
1 MAIN BUCK-BOOST / MEASUREMENT / RECOVERY

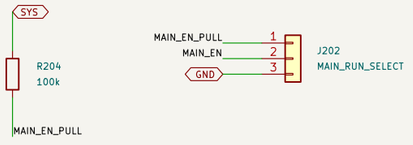


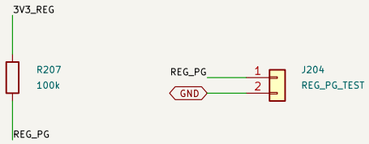




Never parallel an external 3.3 V source.
Open R203 before external injection.
C202 + R201 remain upstream of R203.



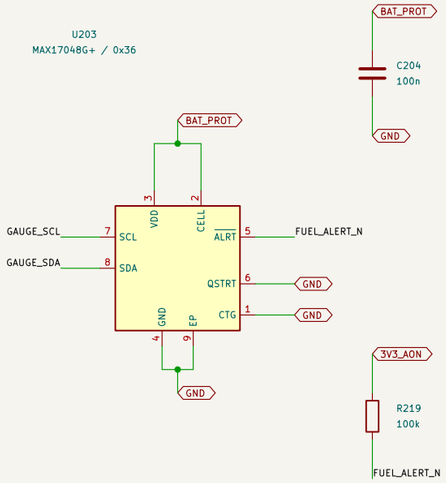




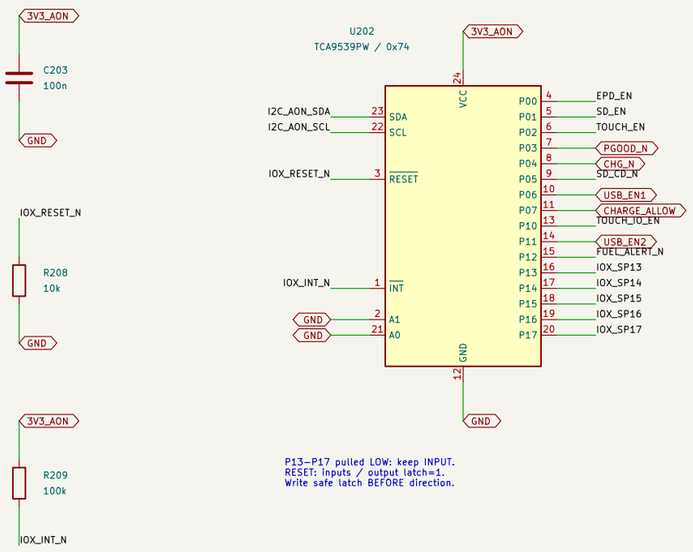
J202: 1-2 RUN; 2-3 OFF / recovery.
One shunt only. Open defaults OFF.

MODE low = automatic PFM/PWM.
U201 exact DLA footprint not assigned.
PG is a test signal, not a new MCU GPIO.

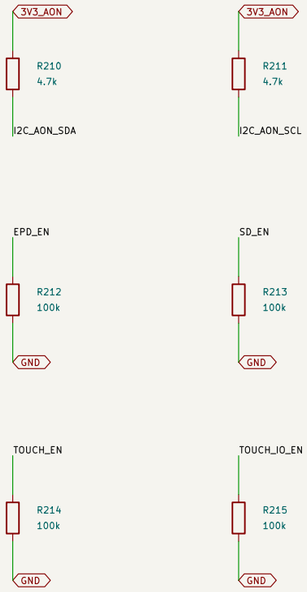
3 BATTERY GAUGE

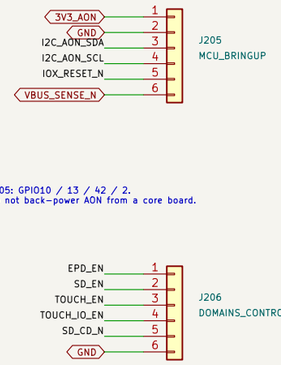


2 AON I2C CONTROL / RESET DEFAULTS



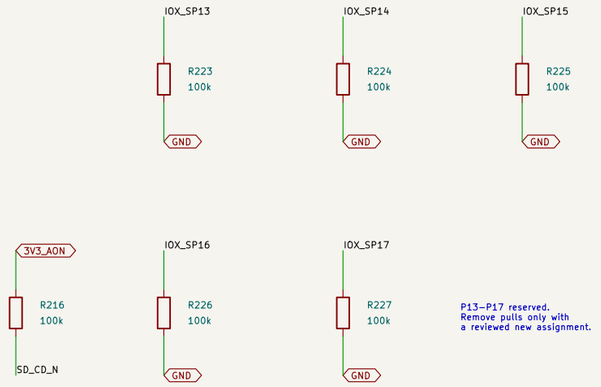
P13-P17 pulled LOW: keep INPUT.
RESET: inputs / output latch=1.
Write safe latch BEFORE direction.





J205: GPIO10 / 13 / 42 / 2.
Do not back-power AON from a core board.

4 SPARE PORT DEFAULTS



P13-P17 reserved.
Remove pulls only with
a reviewed new assignment.

Load switches, TMUX1511, EPD/SD/audio and MCU are future sheets.
Start I2C at 100kHz; edge timing / gauge power-off leakage to verify.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_3V3_CONTROL/
File: POWER_3V3_CONTROL.kicad_sch

Title: 3.3 V REGULATOR / POWER CONTROL / GAUGE

Size: A3	Date: 2026-11-01	**Rev: r0 DRAFT**
KiCad E.D.A. 10.0.7		Id: 3/3